Madelon Hulsebos

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I'm a computer scientist and AI researcher working on the democratization of insights from tabular data (e.g. relational tables and spreadsheets) through table representation learning (TRL). My research involves machine learning, natural language processing, information retrieval and data management.

Employment

Centrum Wiskunde & Informatica (CWI) – Amsterdam, Netherlands
Research Faculty, PI, (tenured in spring 2026), lead of the TRL Lab 2024–present

University of California, Berkeley – Berkeley, United States
Postdoctoral Scholar, EECS and BIDS 2023–2024

University of Amsterdam – Amsterdam, Netherlands
PhD Researcher, Informatics Institute 2023
Guest Researcher, Informatics Institute 2020–2023

Sigma Computing – San Francisco, United States
Full-time PhD Researcher, previously Research Intern (Summer 2021) 2021–2022

KPN/Heineken – Rotterdam/Amsterdam, Netherlands
Data Scientist 2019–2021

Massachusetts Institute of Technology – Cambridge, United States
Research Visitor, MIT Media Lab 2018–2019

Delft University of Technology – Delft, Netherlands
Graduate TA, Pattern Recognition & Web Information Systems groups 2017–2018

Aalto University – Helsinki, Finland
Research and Teaching Assistant, Machine Learning for Big Data 2017

Education

University of Amsterdam – Amsterdam, Netherlands
PhD Computer Science 2020–2024

Delft University of Technology – Delft, Netherlands
MSc Computer Science 2016–2018
BSc Technology, Policy and Management 2011–2015

Ancillary Activities

Prior Labs – Berlin/Freiburg, Germany
Scientific Advisory Board Member 2025–present

ELLIS unit Amsterdam – Amsterdam, Netherlands
Management Team Member 2026-present

UniPartners Delft – Delft, Netherlands
Supervisory Board Member 2017–2023
Executive Board Member 2015–2016

Academic Service

Chairing
Founder and co-chair, Table Representation Learning workshop (TRL) @ **NeurIPS, ACL** 2022-
Co-chair, AI for Tabular Data workshop @ **EurIPS** 2025
Co-chair Tutorial Track @ **VLDB** 2025
Co-chair, New Researcher Symposium @ **SIGMOD** 2025

Last updated: 6 June, 2026

Co-chair Seminar on Challenges and Opportunities in Table Representation Learning @ **Dagstuhl** 2025
Co-chair Representation Learning and Generative Models for Structured Data @ **ELLIS Amsterdam** 2025
Initiator, Table Representation Learning research theme @ **ELLIS Amsterdam** 2025
Co-chair, Data Management for End-to-End ML workshop (DEEM) @ **SIGMOD** 2023-2025
Steering Committee, Tabular Data Analysis workshop (TaDA) @ **VLDB** 2023, 2024
Co-chair, SemTab challenge @ **ISWC** 2021-2023

Conference/Journal Reviewing
ACL 2026
NeurIPS 2026
Conference of Innovative Data Systems Research (CIDR) 2025-2026
SIGMOD 2026
VLDB Journal 2025-2026
NeurIPS D&E track 2022-2025
PVLDB 2024-2026
ICDE (Industry track) 2024
EDBT (Industry track) 2022-2023
TheWebConf (Industry track) 2022-2023

Workshop Reviewing
BioDMS@VLDB'26, NOVAS @ SIGMOD'25, aiDM @ SIGMOD'24, DMLR @ICML'23-'24, DBML @ ICDE'23-'24, PhD workshop @ VLDB'23, SemTab @ ISWC'21-'23, AIDB @ VLDB'22.

Editorial
Assistant Editor, Journal of Systems Research (JSys) 2022 - 2023

Advising and Committees

Advising postdoc/PhD
Dennis Loevlie, CWI, PhD 2026-
Rohith Prabakaran, CWI, PhD 2026-
Aécio Santos, CWI, Postdoc 2025-
Xue (Effy) Li, CWI, Postdoc 2025-
Cornelius Wolff, CWI, PhD 2025-
Daniel Gomm, CWI, PhD 2024-

Advising (under)graduate
Hongqian Xia, University of Amsterdam, MSc 2026
Lisa van Oosten, University of Amsterdam, MSc 2026
Yme Kingma, Radboud University, MSc 2026
Louis Gehringer, University of Amsterdam, MSc 2026
Ahmed Omar Eissa, Utrecht University, MSc 2025-2026
Daniel De Dios Allegue, TU Delft, pre-MSc 2025-2026
Wojciech Kosiuk, University of Amsterdam, MSc 2025-2026
Jan Henrik Bertrand, University of Amsterdam, MSc 2025-2026
Liang Telkamp, University of Amsterdam, MSc (🏆Amsterdam AI Thesis Award) 2025
Nikola Selic, TU Munich, MSc 2025
Wenjing Lin, UC Berkeley, MSc 2024-2026
Rachel Lin, UC Berkeley, MSc 2024-2025
Xingyu Ji, UC Berkeley, BSc 2023-2025
Teun Mathijssen, University of Amsterdam, MSc 2023

Thesis Committees
Irina Saparina, University of Edinburgh, PhD 2026
Lukas Lakowski, HPI, PhD 2026
Tanguy Urvoy, University of Renness, HDR 2026
Lennart Behme, BIFOLD/TU Berlin, PhD 2026
Félix Lefebvre, Inria, PhD 2026
Margherita Martorana, VU Amsterdam, PhD 2025
Vaishali Pal, University of Amsterdam, PhD 2025

Tianji Cong, University of Michigan, PhD 2023-2025

Awards and Funding (€2.3M+)

Compute/model credit grant, $35K, Google 2026
Table Representation Learning Theme at ELLIS unit Amsterdam, €50K, SAP 2026
Research grant for collaboration CWI x Inria on TRL, €727K, CWI 2025
Model credit grants, $7K, OpenAI & Cohere 2025
Amsterdam AI Thesis Award for the supervised MSc thesis by Liang Telkamp 2025
Table Representation Learning Theme at ELLIS unit Amsterdam, €50K, SAP 2024
Table Representation Learning Research Grant, €300K, SAP 2024
AiNed Fellowship Grant, €993K, NWO 2024
ELLIS Membership 2024
Postdoctoral Fellowship, $150K, Accenture-BIDS 2023
Best Reviewer Award, PhD Workshop, VLDB 2023
Travel Award, $2.5K, VLDB Endowment 2022
Honorable mention GitTables, SemTab challenge 2021

Talks, Panels and Lectures

Invited Talks and Keynotes

What AI is Leaving on the Table [keynote]
Deep Learning Extravaganza, ELLIS Amsterdam 2026
AI for Good Webinar "Discovery: From Molecules to Models", ELLIS & UN ITU 2026

Towards Contextual Sensitive Data Detection
ICMD Consultation, Raoul Wallenberg Institute, Sweden 2026
UNECE Expert Meeting on Statistical Data Confidentiality, Spain 2025

What are we asking from Tabular Data? [keynote]
AI for Tabular Data workshop, EurIPS, Denmark 2025

Towards robust open-domain querying over structured data
KIK AI seminar, Amsterdam University Medical Center, Netherlands 2025
ELLIS NL Scientific Day, Netherlands 2025
AI Research Retreat, SAP, Germany 2025
Business Analytics Seminar, University of Amsterdam, Netherlands 2025
Seminar on AI, Semantics, and Tabular Data, IBM, USA 2025

What Table Representation Learning brings to Data Systems
COMPASS Seminar, ETH, Zürich, Switzerland 2024
CWI Lectures, CWI, Amsterdam, Netherlands 2024

Retrieval Systems for Structured Data: the missing piece for grounding LLM-powered query interfaces in factual data
Berkeley Institute for Data Science Seminar, Berkeley, USA 2024
Berkeley EECS EPIC Retreat, Berkeley, USA 2024
Harnessing AI for Relational Data: Industry and Research Perspectives, SF, USA 2024

Advances, challenges, and opportunities in Table Representation Learning
Microsoft Gray Systems Lab, USA 2024
University of Washington Seattle, Seattle, USA 2024
Snowflake, USA 2024
Google, Sunnyvale, USA 2024
UC Berkeley, Berkeley, USA 2024
Transformers at Work 2023, Zeta Alpha, Amsterdam, Netherlands 2023

Towards Table Representation Learning for end-to-end data management and analysis
INRIA-Saclay, Paris, France 2023
ML for Systems and Systems for ML Workshop @ BTW, Dresden, Germany 2023
Hasso Plattner Institute, Berlin, Germany 2023
TU Darmstadt, Darmstadt, Germany 2022

GitTables: a large corpus of relational tables
Tabular Data Analysis workshop, VLDB, Vancouver, Canada 2023

Database Architectures group, CWI, Amsterdam, Netherlands 2022

(Guest) Lectures
Representation Learning and Generative Models for Tabular Data
University of Amsterdam 2026
BIFOLD, TU Berlin 2025

Panel discussions
"Neural Relational Data: Tabular Foundation Models, LLMs... or both?", VLDB, London 2025
Transformers at Work 2023, Zeta Alpha, Amsterdam, Netherlands 2023

Publications

2026

SQaLe: A Large Dataset for Specialized Text-to-SQL, **Under Review**
Wolff, C., Gomm, D., Hulsebos, M.

Fine-Grained Table Retrieval Through the Lens of Complex Queries, **Under Review**
Kosiuk, W., Ji, X., Chung, Y., Özcan, F., Hulsebos, M.

Towards Contextual Sensitive Data Detection, **Under Review**
Telkamp, L., Hulsebos, M.

2025

SQaLe: A Large Text-to-SQL Corpus Grounded in Real Schemas, **AI for Tabular Data workshop at EurIPS 2025**
Wolff, C., Gomm, D., Hulsebos, M.

Are We Asking the Right Questions? On Ambiguity in Natural Language Queries for Tabular Data Analysis, **AI for Tabular Data workshop at EurIPS 2025**
Gomm, D., Wolff, C., Hulsebos, M.

Unlocking the Full Potential of Data Science Requires Tabular Foundation Models, Agents, and Humans
Cong, T., Eisenschlos, J., Gomm, D., Grinsztajn, L., Müller, A., Sanghi, A, et al., Binnig, C., Hulsebos, M., Hutter, F.

How well do LLMs reason over tabular data, really?, **Table Representation Learning workshop at ACL**
Wolff, D., Hulsebos, M.

Rethinking Dataset Discovery with DataScout, **User Interface Software and Technology (UIST)**
Lin, R., Chopra, B.., Lin, W. Shankar, S. Hulsebos, M., Parameswaran, A.

Metadata Matters in Dense Table Retrieval, **ELLIS workshop on RL and GM for Structured Data**
Gomm, D., Hulsebos, M.

TARGET: Benchmarking Table Retrieval for Generative Tasks, **In Review**
Li, X., Glenn, P., Parameswaran, A., Hulsebos, M.

*Towards Accurate and Efficient Document Analytics with Large Language Models***, ICDE**
Lin, Y., Hulsebos, M., Ma, R., Shankar, S., Zeigham, S., Parameswaran, A. G., & Wu, E.

2024

TARGET: Benchmarking Table Retrieval for Generative Tasks, **TRL @ NeurIPS**
Li, X., Parameswaran, A., Hulsebos, M.

"It Took Longer than I was Expecting:" Why is Dataset Search Still so Hard?, **HILDA@SIGMOD**
Hulsebos, M., Lin, W., Shankar, S., Parameswaran, A.

SchemaPile: A Large Collection of Relational Database Schemas, **SIGMOD**
Doehmen, T., Geacu, R., Hulsebos, M., Schelter, S.

SPADE: Synthesizing Assertions for Large Language Model Pipelines, **Proceedings of VLDB**

Shankar, S., Li, H., Asawa, P., Hulsebos, M., Lin, Y., Zamfirescu-Pereira, J., Chase, H., Fu-Hinthorn, W., Parameswaran, A., Wu, E.

Eighth Workshop on Data Management for End-to-End Machine Learning (DEEM), **SIGMOD**
Hulsebos, M., Shankar, S., Interlandi, M.

2023
AdaTyper: Adaptive Semantic Type Detection, **ArXived**
Hulsebos, M., Groth, P., Demiralp, C.

Introducing the Observatory Library for End-to-End Table Embedding Inference, **TRL @ NeurIPS**
Cong, T., Sun., Z., Groth, P., Jagadish, H., Hulsebos, M.

Observatory: Characterizing Embeddings of Relational Tables, **Proceedings of VLDB**
Cong, T., Hulsebos, M., Sun., Z. Groth, P., Jagadish, H.V.

Models and Practice of Neural Table Representations [tutorial], **SIGMOD**
Hulsebos, M., Deng, X., Sun, H., Papotti, P.

Seventh Workshop on Data Management for End-to-End Machine Learning (DEEM), **SIGMOD**
Boehm, M., Hulsebos, M., Shankar, S., Varma, P.

2022
GitTables: A Large-Scale Corpus of Relational Tables, **SIGMOD**
Hulsebos, M., Demiralp, C., Groth, P.

GitSchemas: A Dataset for Automating Relational Data Preparation Tasks, **DBML @ ICDE**
Döhmen, T., Hulsebos, M., Beecks, C., Schelter, S.

Results of SemTab 2022, **Proceedings of SemTab @ ISWC**
Abdelmageed, N., Chen, J., Cutrona, V., Efthymiou, V., Hassanzadeh, O., Hulsebos, M., Jiménez-Ruiz, E., Sequeda, J. and Srinivas, K.

Making Table Understanding Work in Practice [abstract], **CIDR**
Hulsebos, M., Gathani, S., Gale, J., Dillig, I., Groth, P., Demiralp, C.

Augmenting Decision Making via Interactive What-If Analysis, **CIDR**
Gathani, S., Hulsebos, M., Gale, J., Haas, P. J., Demiralp, C.

2021
Results of SemTab 2021, **Proceedings of SemTab @ ISWC**
Cutrona, V., Chen, J., Efthymiou, V., Hassanzadeh, O., Jiménez-Ruiz, E., Sequeda, J., Srinivas, K., Abdelmageed, N., Hulsebos, M., Oliveira, D., Pesquita, C.

2020
Sato: Contextual semantic type detection in tables, **Proceedings of VLDB**
Zhang, D., Suhara, Y., Li, J., Hulsebos, M., Demiralp, C., Tan, W.

2019
Sherlock: A deep learning approach to semantic data type detection, **ACM SIGKDD**
Hulsebos, M., Hu, K., Bakker, M., Zgraggen, E., Satyanarayan, A., Kraska, T., Demiralp, C., Hidalgo, C.

VizNet: Towards a large-scale visualization learning and benchmarking repository, **ACM CHI**
Hu, K., Gaikwad, N., Hulsebos, M., Bakker, M., Zgraggen, E., Hidalgo, C., Kraska, T., Li, G., Satyanarayan, A., Demiralp, C.

2018
The Network Nullspace Property for Compressed Sensing of Big Data Over Networks, **IEEE ICASSP**
Hulsebos, M., Jung, A.